

BOSTON BUOY 44013, MA, USAWMO: **992420**Lat: **42.346N**Lon: **70.651W**Elev: **0**StdP: **101.33**Time Zone: **-5.00 (NAE)**Period: **94-18**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-11.3	-8.9	-17.7	0.8	-9.2	-16.2	0.9	-7.9	17.7	-3.5	16.5	-3.7	10.4	310	0.640

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	3.5	24.4	21.0	23.2	20.5	22.3	20.0	22.1	23.1	21.4	22.4	20.7	21.8	4.3	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
21.8	16.5	22.4	21.0	15.6	21.9	20.3	15.0	21.4	64.8	23.2	62.1	22.5	59.8	21.9	25.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 15.4	13.9	11.8	DB	-14.3	26.8	2.5	1.3	-16.1	27.7	-17.5	28.5	-19.0	29.2	-20.8	30.2	(4)
(5)			WB	-15.3	23.4	2.6	1.7	-17.2	24.6	-18.8	25.5	-20.2	26.5	-22.2	27.7	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	9.8	-0.3	0.1	2.7	6.9	11.4	16.2	19.8	19.7	17.2	12.5	7.8	3.5
	DBStd	7.75	4.93	4.02	3.26	2.73	2.63	2.79	1.92	1.85	2.22	2.79	3.49	4.04
	HDD10.0	1230	319	277	227	99	13	0	0	0	0	10	81	204
	HDD18.3	3226	578	510	484	343	214	74	8	8	48	182	316	461
	CDD10.0	1173	0	0	1	6	58	186	303	301	216	87	15	2
	CDD18.3	127	0	0	0	0	1	11	52	50	13	1	0	0
	CDH23.3	86	0	0	0	0	1	6	45	29	5	0	0	0
	CDH26.7	1	0	0	0	0	0	0	1	1	0	0	0	0
Wind	WSAvg	6.0	8.0	7.5	6.9	5.7	4.9	4.3	4.0	4.2	5.0	6.4	7.1	7.9
Precipitation	PrecAvg	1290	107	95	138	113	101	93	89	103	105	117	105	120
	PrecMax	1544	179	202	412	236	313	320	216	165	221	323	187	209
	PrecMin	995	38	51	11	29	31	3	40	24	26	16	24	38
	PrecStd	166	37	42	87	55	62	80	43	45	58	74	47	51
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.1	10.2	12.5	16.3	20.6	24.1	26.0	25.7	24.1	20.2	16.6	14.1
		MCWB	10.6	8.6	8.6	12.6	16.9	19.6	21.9	21.7	21.6	17.5	15.1	12.3
	2%	DB	9.2	7.8	9.6	13.7	18.1	22.2	24.6	24.1	22.3	18.3	14.6	11.6
		MCWB	8.0	6.3	7.3	10.9	15.3	18.7	21.1	21.0	20.3	16.1	12.8	10.2
	5%	DB	7.6	6.2	8.0	11.7	16.5	20.9	23.4	23.0	21.2	17.0	13.4	10.3
		MCWB	6.1	4.6	6.0	9.4	14.0	18.2	20.6	20.6	19.5	15.2	11.6	9.2
	10%	DB	6.1	5.1	6.6	10.3	15.0	19.8	22.4	22.2	20.2	16.1	12.4	9.0
		MCWB	4.7	3.6	4.9	8.2	13.1	17.6	20.1	20.2	18.7	14.4	10.8	8.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	10.2	8.8	9.6	13.8	17.9	20.8	23.6	23.0	23.0	18.6	15.7	12.7
		MCDB	10.5	9.4	10.3	15.9	19.6	22.7	24.7	23.8	23.4	19.7	16.3	13.2
	2%	WB	7.8	6.3	7.6	11.9	16.1	19.7	22.0	22.1	21.4	17.0	13.4	10.7
		MCDB	8.5	7.0	8.8	13.7	17.3	21.3	23.2	23.1	22.0	17.8	14.2	11.4
	5%	WB	5.9	4.8	6.3	10.2	14.8	18.8	21.3	21.4	20.2	16.0	12.1	9.5
		MCDB	6.7	5.6	7.3	11.5	15.5	20.2	22.5	22.4	20.9	16.8	12.9	10.1
	10%	WB	4.3	3.5	5.2	8.9	13.4	18.0	20.6	20.7	19.3	15.1	11.1	8.1
		MCDB	5.5	4.5	6.4	10.2	14.4	19.3	21.9	21.7	20.1	15.8	12.2	8.8
Mean Daily Temperature Range	5% DB	MDBR	5.4	5.0	4.4	3.9	4.0	3.9	3.8	3.5	3.3	3.8	4.1	4.8
		MCDBR	6.5	6.0	6.0	6.5	6.7	5.9	5.3	4.7	4.7	4.8	5.0	5.7
		MCWBR	6.6	6.0	4.7	4.7	4.7	3.6	3.1	3.1	3.6	3.9	5.0	5.8
	5% WB	MCDBR	6.1	5.6	5.5	6.5	6.2	5.2	4.6	4.2	4.2	4.1	4.7	5.6
		MCWBR	6.7	6.0	5.0	5.2	4.9	3.6	3.2	3.3	3.7	4.0	5.3	6.3
Clear-Sky Solar Irradiance	taub	0.298	0.305	0.325	0.369	0.405	0.455	0.475	0.435	0.378	0.338	0.313	0.299	
	taud	2.525	2.515	2.506	2.407	2.349	2.245	2.196	2.296	2.439	2.549	2.577	2.575	
	Ebn at Noon	856	907	924	900	871	825	803	825	854	856	832	826	
	Edh at Noon	70	83	94	112	122	136	142	124	100	80	67	62	
All-Sky Solar Radiation	RadAvg	1.71	2.59	3.61	4.67	5.24	5.65	5.94	5.34	4.28	2.86	1.91	1.42	
	RadStd	0.12	0.23	0.25	0.50	0.56	0.52	0.36	0.22	0.28	0.24	0.14	0.16	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (68 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+93	+42

Nomenclature: See separate page